

Lab Experiment: Exploring Hugging Face – From Manual ML to Pretrained Models

Course Outcome Focus

- Understand ML concepts manually
- Implement ML using code
- Explore modern ML using pretrained models via **Hugging Face**

Question

You are given a small dataset for a **binary classification problem** (e.g., Spam Detection or Sentiment Analysis).

Task 1: Build a Model from Scratch

1. Choose a suitable **machine learning algorithm** (Decision Tree / Naïve Bayes / Logistic Regression).
2. Solve the problem **manually** using a dataset of **minimum 10 data points**.
3. Implement the same model using **Python code from scratch** (without using pretrained models).

Task 2: Solve the Same Problem Using a Pretrained Model

1. Use a **pretrained model from Hugging Face** to solve the same classification problem.
2. Compare the results with the manually solved and coded model.
3. Comment on performance, ease of use, and accuracy.

Sample Dataset (10 Data Points – Sentiment Analysis)

ID	Text	Sentiment
1	I love this product	Positive
2	Very bad experience	Negative
3	Amazing quality	Positive
4	Not worth the money	Negative
5	Excellent performance	Positive
6	Terrible support	Negative
7	Happy with purchase	Positive
8	Waste of time	Negative
9	Good value	Positive
10	Disappointed	Negative

◇ Part A: Manual Solution (Conceptual) – This has to be in Observation

Chosen Algorithm: Naïve Bayes (Text Classification)

Steps (Manual):

1. Count word frequencies for **Positive** and **Negative** classes.
2. Compute prior probabilities:
 - $P(\text{Positive}) = 5/10$
 - $P(\text{Negative}) = 5/10$
3. For a new sentence "*Excellent product*":
 - Compute likelihood of words under both classes.
 - Multiply likelihood with prior.
4. Assign class with **maximum posterior probability**.

Manual Result:

Sentence classified as **Positive**

◇ Part B: Code-Based Implementation (From Scratch)

Approach:

- Convert text to bag-of-words
- Train Naïve Bayes using `scikit-learn` (no pretrained models)

Outcome:

- Model correctly classifies most test samples
- Accuracy depends heavily on dataset size

◇ Part C: Pretrained Model using Hugging Face

Steps:

1. Load a pretrained sentiment model from Hugging Face
Example: `distilbert-base-uncased-finetuned-sst-2-english`
2. Use Hugging Face `pipeline("sentiment-analysis")`
3. Input same text samples
4. Observe predictions

Output Example:

Text: "Excellent product"
Prediction: POSITIVE (Score: 0.99)